

X4Pro. News-2026.

by V.Zerkin, Vienna, 2026-08-25

1. Code reac1.py: universal retrieval and plotting of EXFOR data having one-string Reaction-codes, for example:

- CSP Partial cross section data
- CST Temperature dependent cross section
- ETA Average neutron yield per nonelastic event
- NU Fission-neutron yield, nu-bar
- TKE Total kinetic energy
- SIG Cross section data given with limit as DATA and DATA-MIN

2. Plotting partial cross sections from local EXFOR and remote ENDF databases

- CSP vs ENDF:MT51:MF3+33 - Production of a neutron, with residual in the 1st excited state
- NU vs ENDF:MT452/455/456:MF1+31 - Average number of total/delayed/prompt neutrons released per fission event

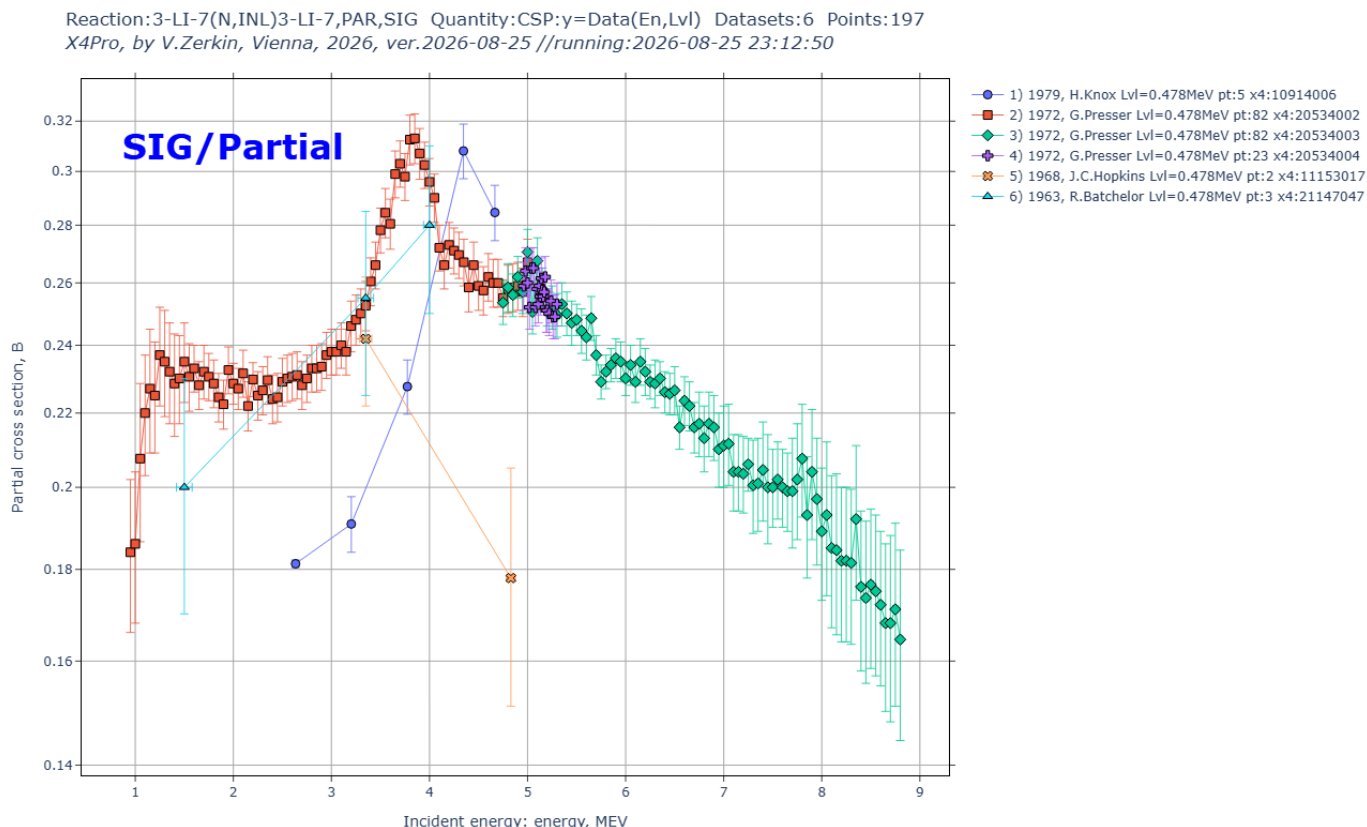
Code reac1.py help:

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Program: reac1.py, ver.2026-08-14
Author: V.Zerkin, Vienna, 2021-2026
Purpose: Retrieve and plot any type EXFOR data from local EXFOR database
Introduction:
* computational data are presented as function y=f(x1,x2,...)
  where f is measured quantity, xi are independent variables:
    EN - incident energy,          E2 - energy of outgoing particle,
    ANG - angle of outgoing particle,  EVL - energy level, etc.
Algorithm:
* user defines EXFOR Reaction-code for data search, x-variable by number
* selects x-variable by number (default: x1)
* other variables will be used as parameter for grouping of data points
Usage:
$ python [{flag}] x4update.py [{option|reaction}]
* flag: see all Python flags: $ python --help
-B      don't write .pyc files on import
* option:
-help    print this help-text and exit (also --help)
-o:<file> output file name
-x:x?    set x variable: "x1" to "x5", e.g. -x:x2, default: -x:x1
-x?fam:<str> filter for "families" of x? variable, e.g. -x2fam:"LVL;EXC"
-x?min:<num> filter out values of x? variable, e.g. -x1min:6e6
-x?max:<num> filter out values of x? variable, e.g. -x1max:30e6
-nogrp   avoid grouping datasets by reaction-codes on the plot
-fx:<num> set factor for x-values, e.g. -fx:1e6 sets "MeV" on X-Axis
-fy:<num> set factor for y-values, e.g. -fy:1e-3 sets "mb/sr" on Y-Axis
plotting:
-xlog    logarithmic scale of X-Axis
-ylog    logarithmic scale of Y-Axis
-lines   connect points on the plot by lines
-sym     draw border of a symbol of a data point on the plot
-annot:x,y,txt annotation on top of the plot;
          x and y should be given in units of the plot;
          text can include html tags: <b>, <sup>
* reaction:
<reaction> full EXFOR Reaction-code, e.g. "8-0-16(N,EL)8-0-16,,DA"
  
```

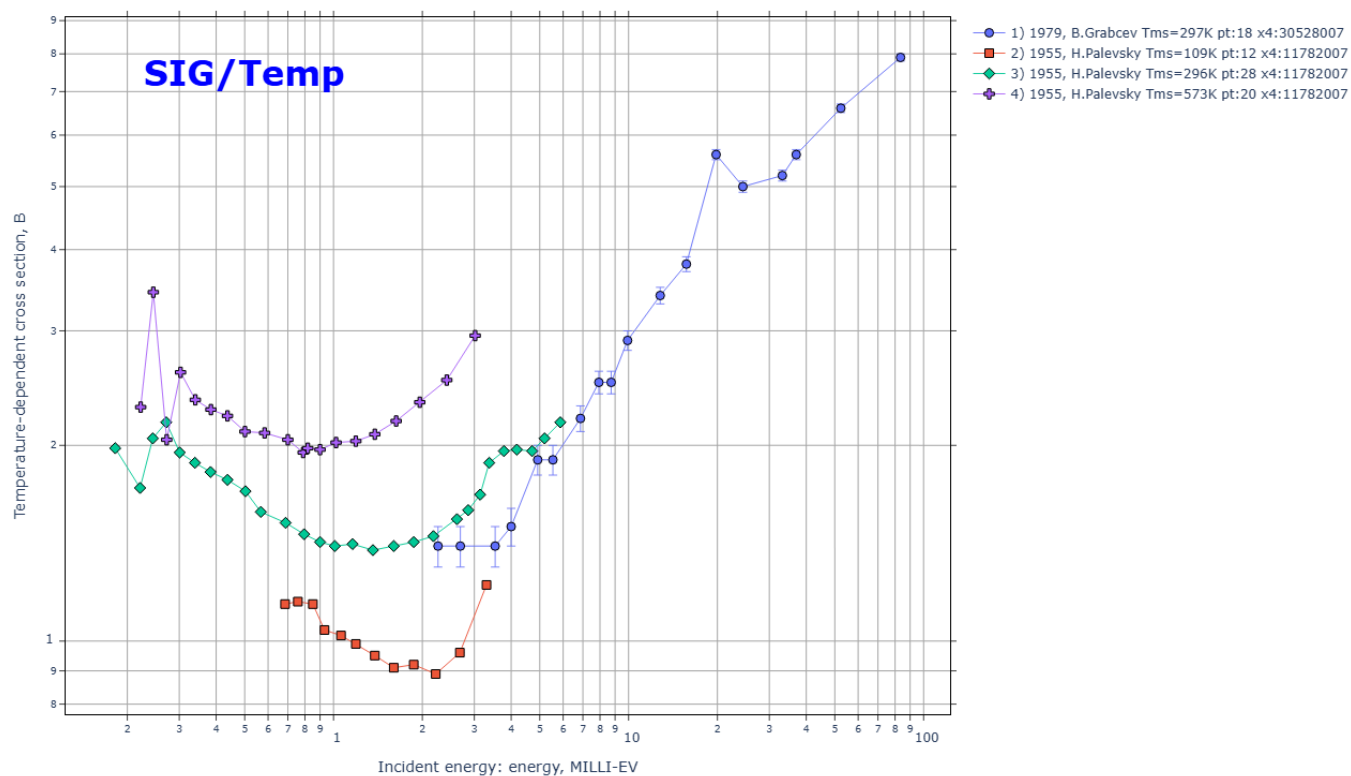
part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

1) part1-8-reac/out00/sig1p.html.png CSP Partial cross section data



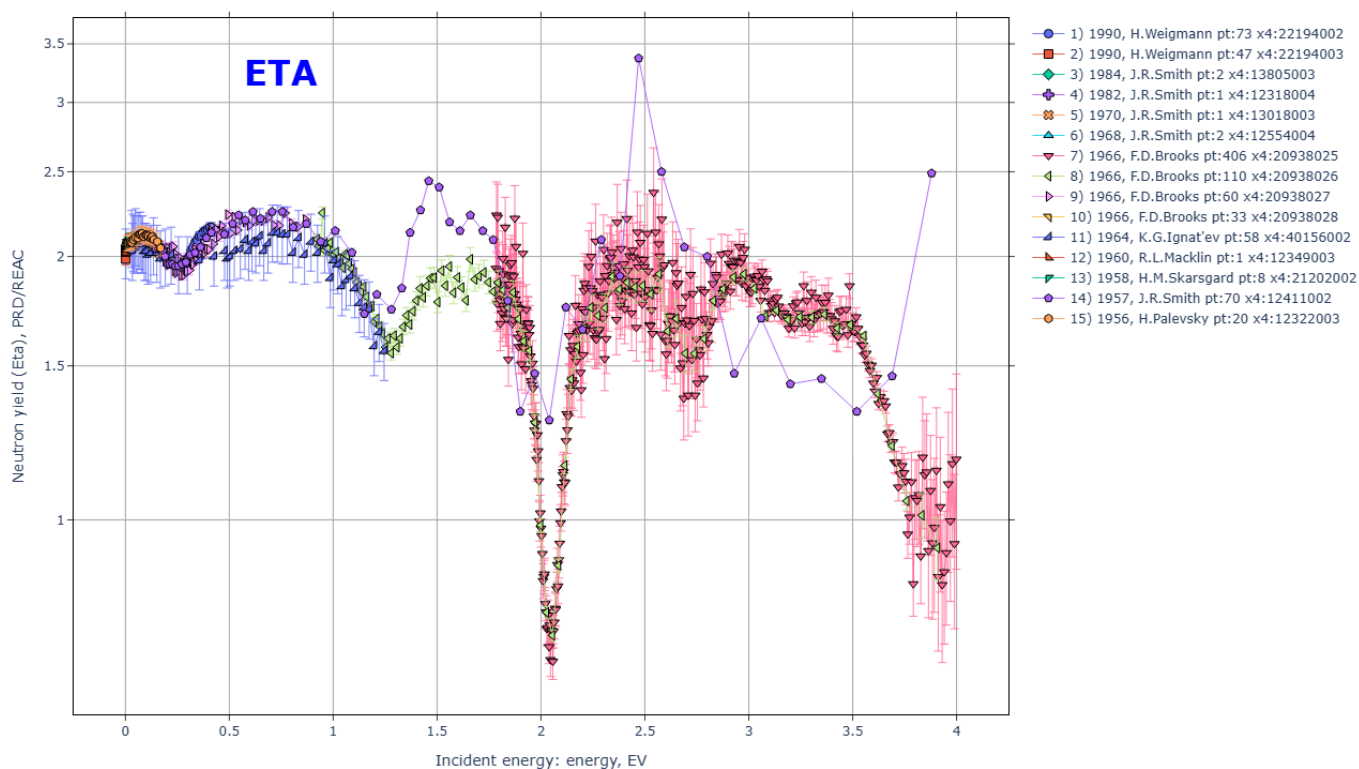
2) part1-8-reac/out00/cst1.html.png CST Temperature dependent cross section

Reaction:82-PB-0(N,TOT),,SIG/TMP Quantity:CST:y=Data(En,Tms) Datasets:4 Points:78
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-25 //running:2026-08-25 23:12:36



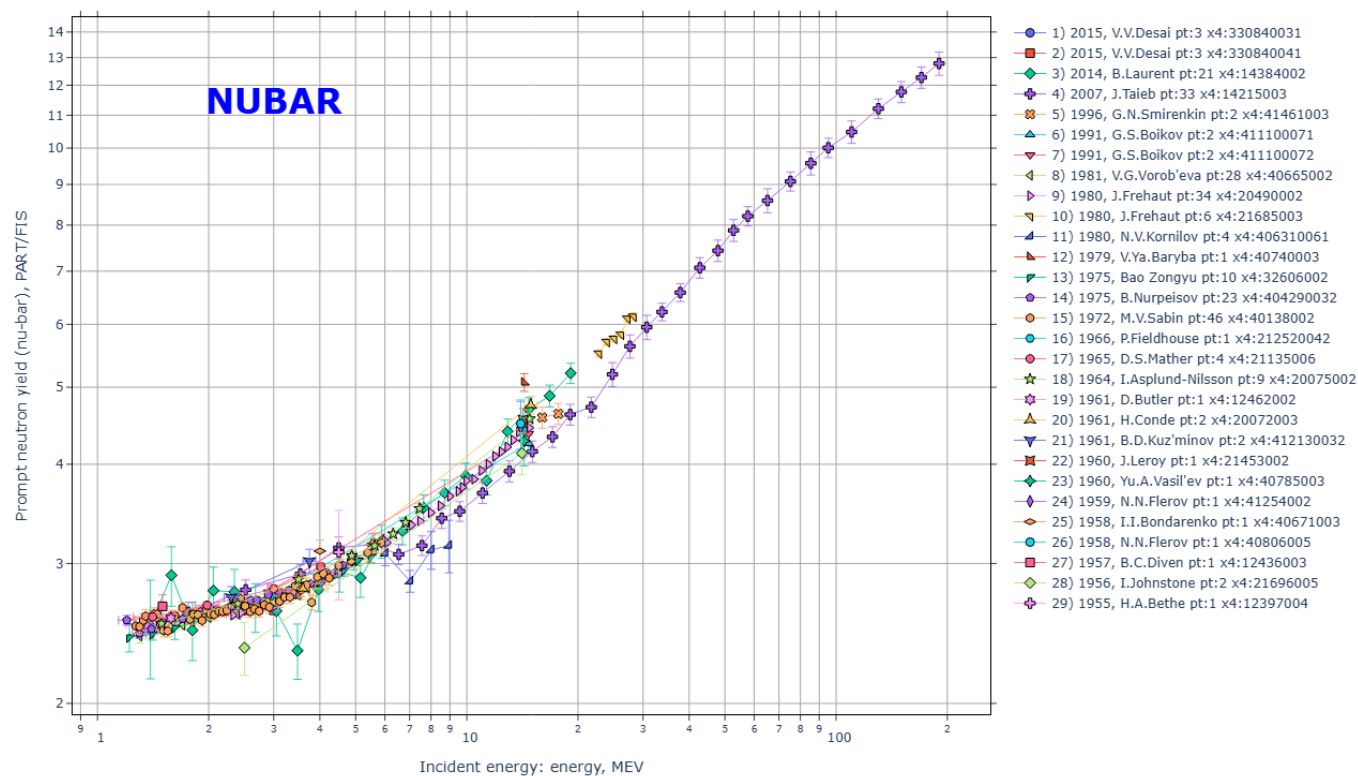
3) part1-8-reac/out00/eta1.html.png ETA Average neutron yield per nonelastic event

Reaction:92-U-235(N,ABS),,ETA Quantity:NU:y=Data(En) Datasets:15 Points:892
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-25 //running:2026-08-25 23:12:46



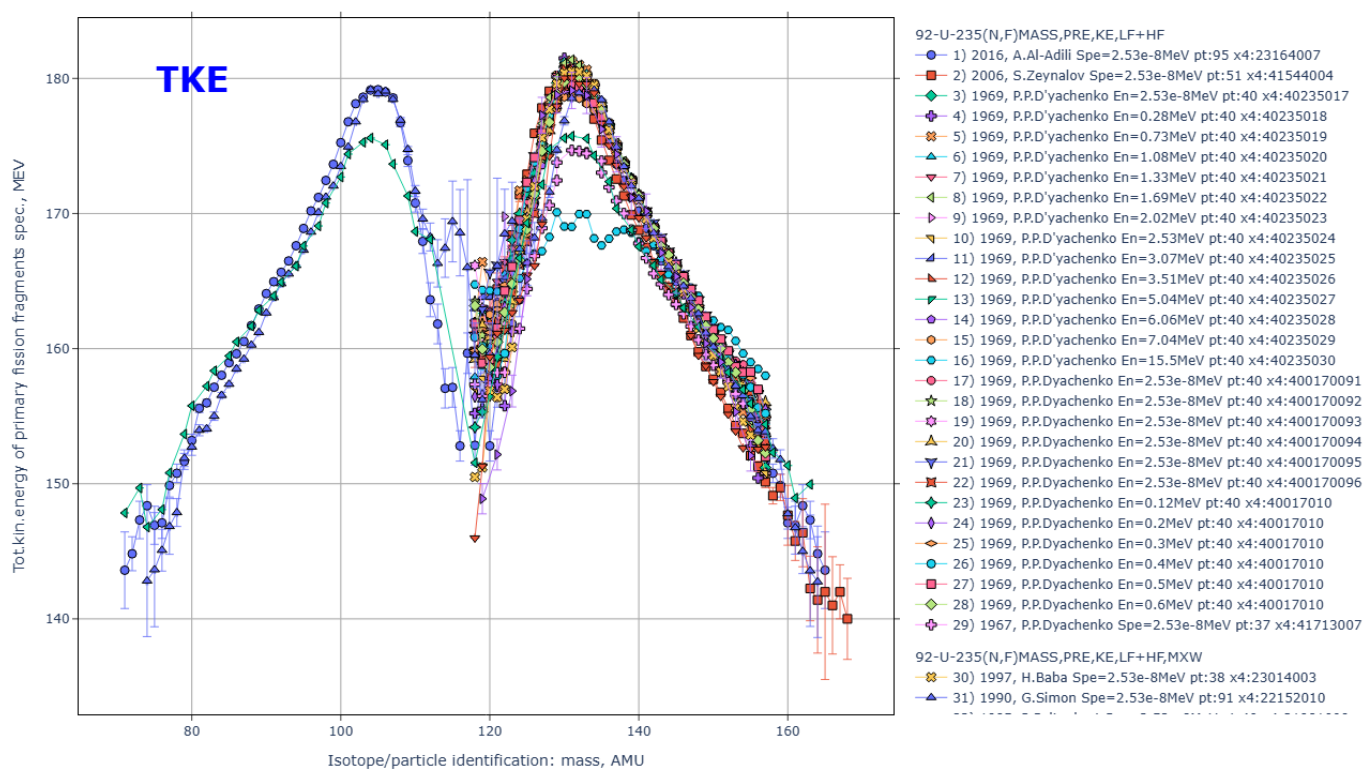
4) [part1-8-reac/out00/nu1.html.png](#) NU Fission-neutron yield, nu-bar

Reaction: 92-U-238(N,F),PR,NU Quantity: NU:y=Data(En) Datasets: 29 Points: 246
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-25 //running:2026-08-25 23:12:40



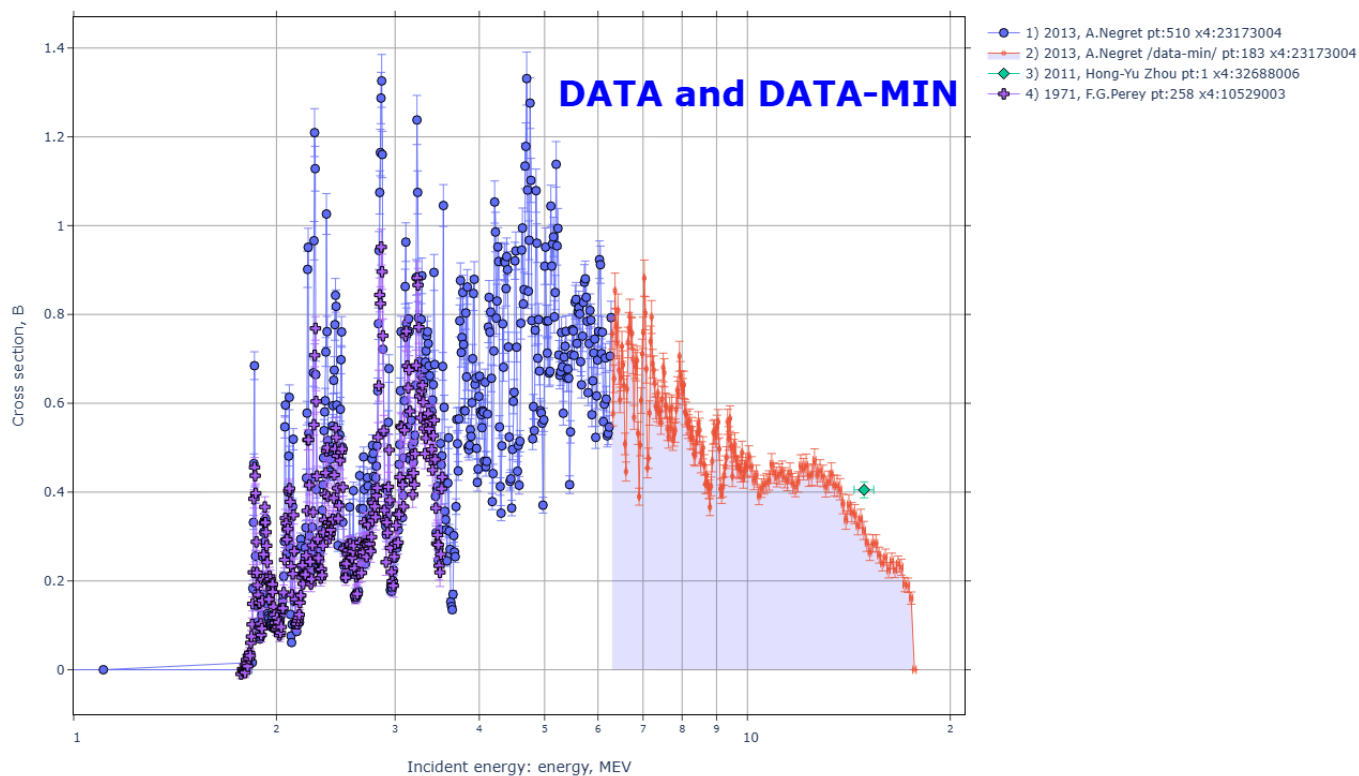
5) [part1-8-reac/out00/tke1.html.png](#) TKE Total kinetic energy

Reaction: 92-U-235(N,F)MASS,PRE,KE,LF+HF Quantity: E:y=Data(Spe,Zam) Datasets: 34 Points: 1467
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-25 //running:2026-08-25 23:12:43



6) [part1-8-reac/out00/si28nn.html.png](#) SIG Cross section data given with limit as DATA and DATA-MIN

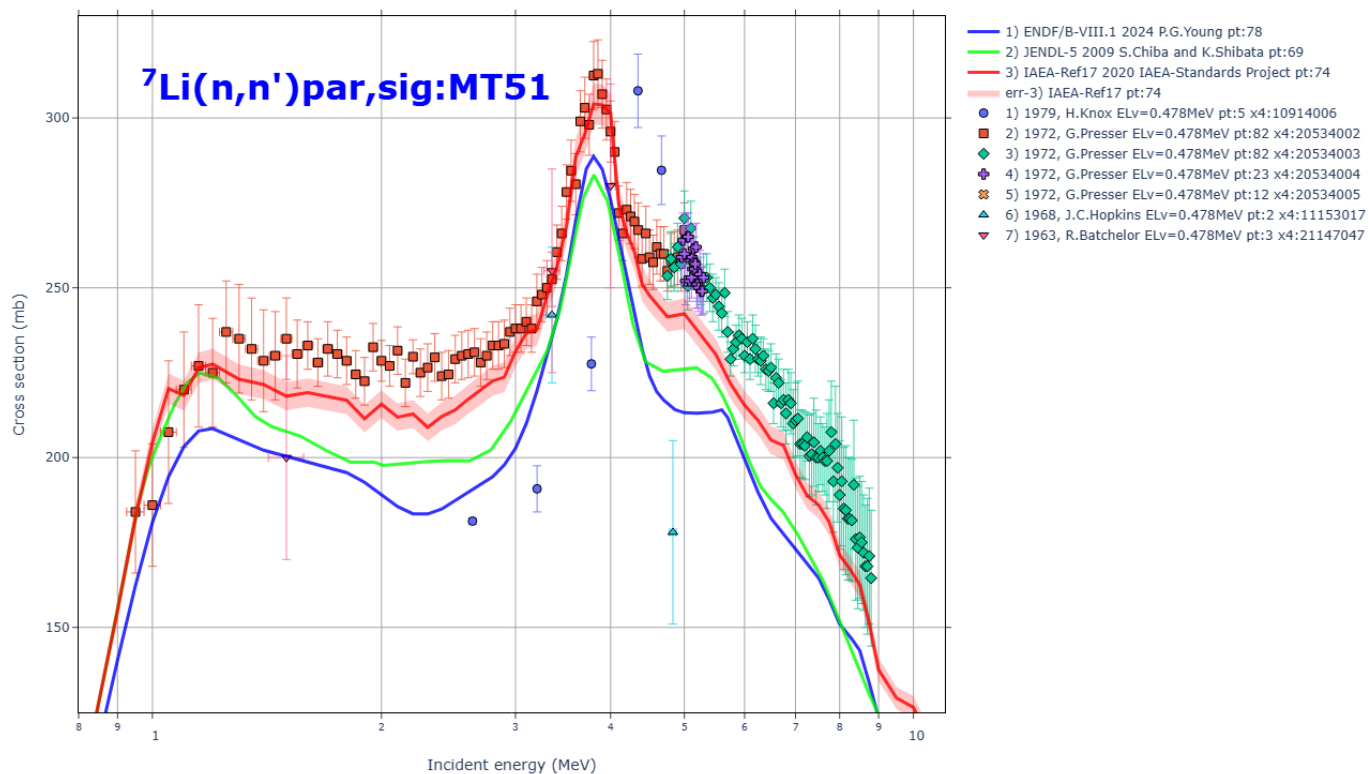
Reaction: 14-SI-28(N,INL)14-SI-28,,SIG Quantity:CS:y=Data(En) Datasets:4 Points:952
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-25 //running:2026-08-25 23:12:53



part2-8-sig1par/ EXFOR/ENDF partial cross sections CSP:MT51

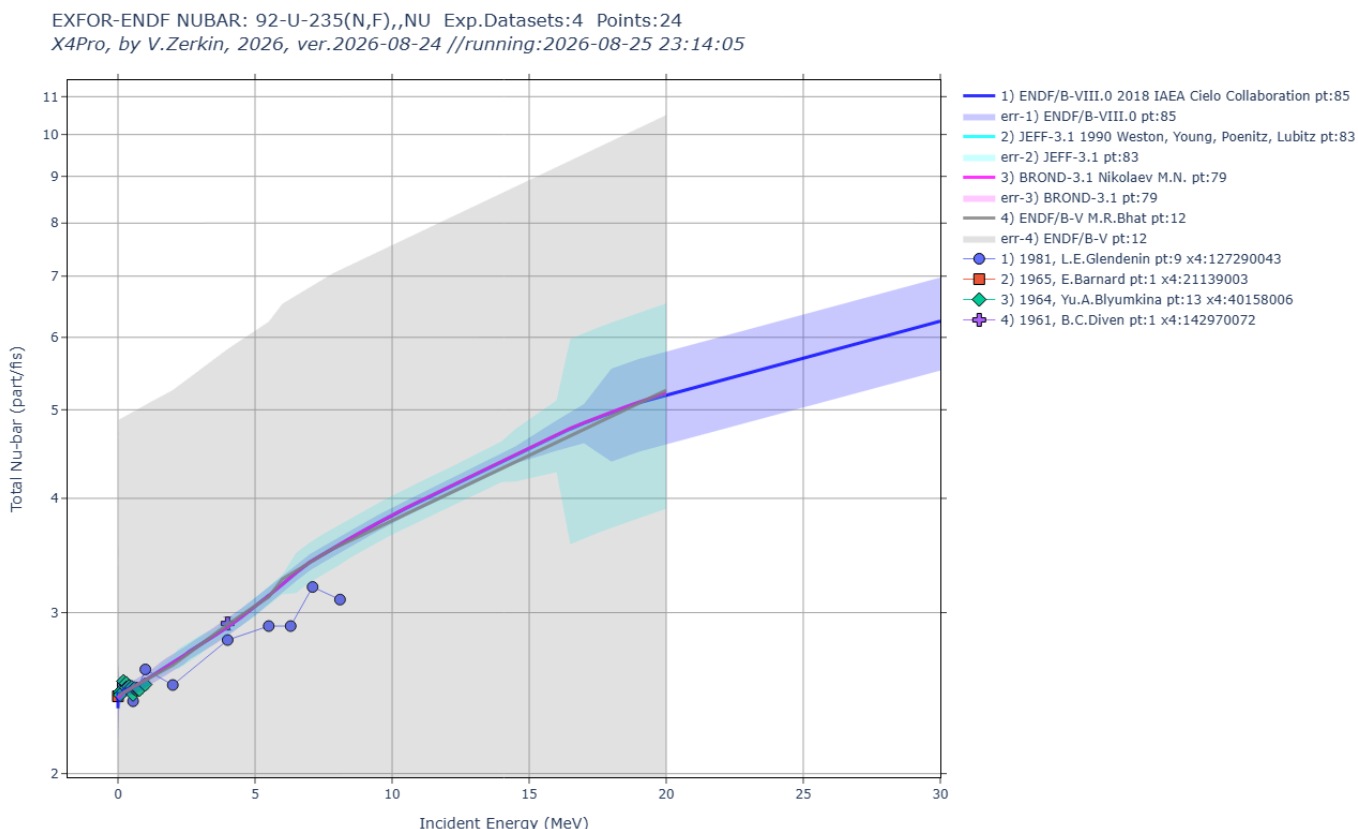
7) [part2-8-sig1par/out00/Li7ninl.html.png](#) Production of a neutron, with residual in the 1st excited state

EXFOR/ENDF cross sections SIG(E): Li-7(n,inl)par,sig EXFOR-datasets:7
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2026, ver.2026-08-10 //run:2026-08-25 23:13:55

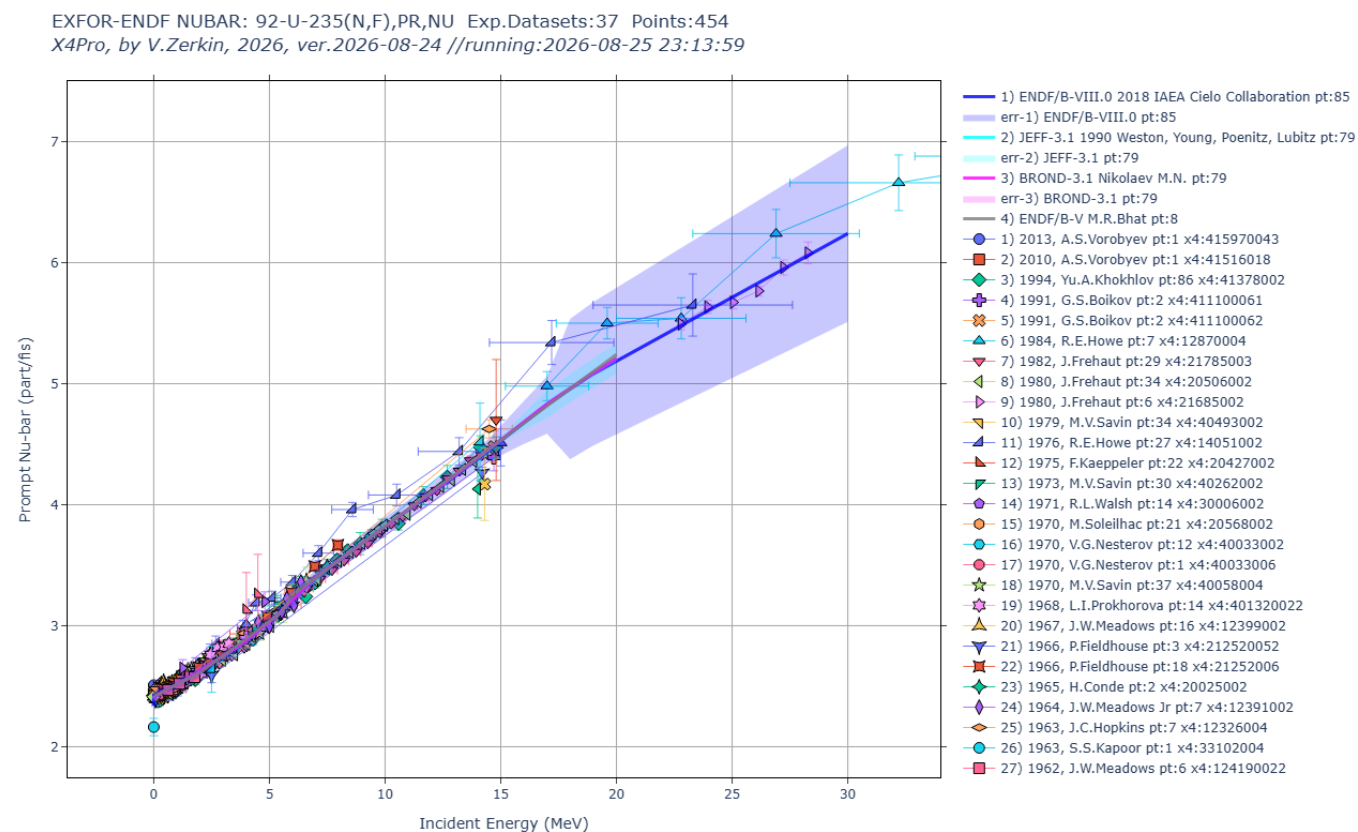


part2-9-nubar/ EXFOR/ENDF NUBAR - average number of neutrons per fission

8) part2-9-nubar/out00/U235nf_-nu.html.png Total neutron yield (nu-bar)

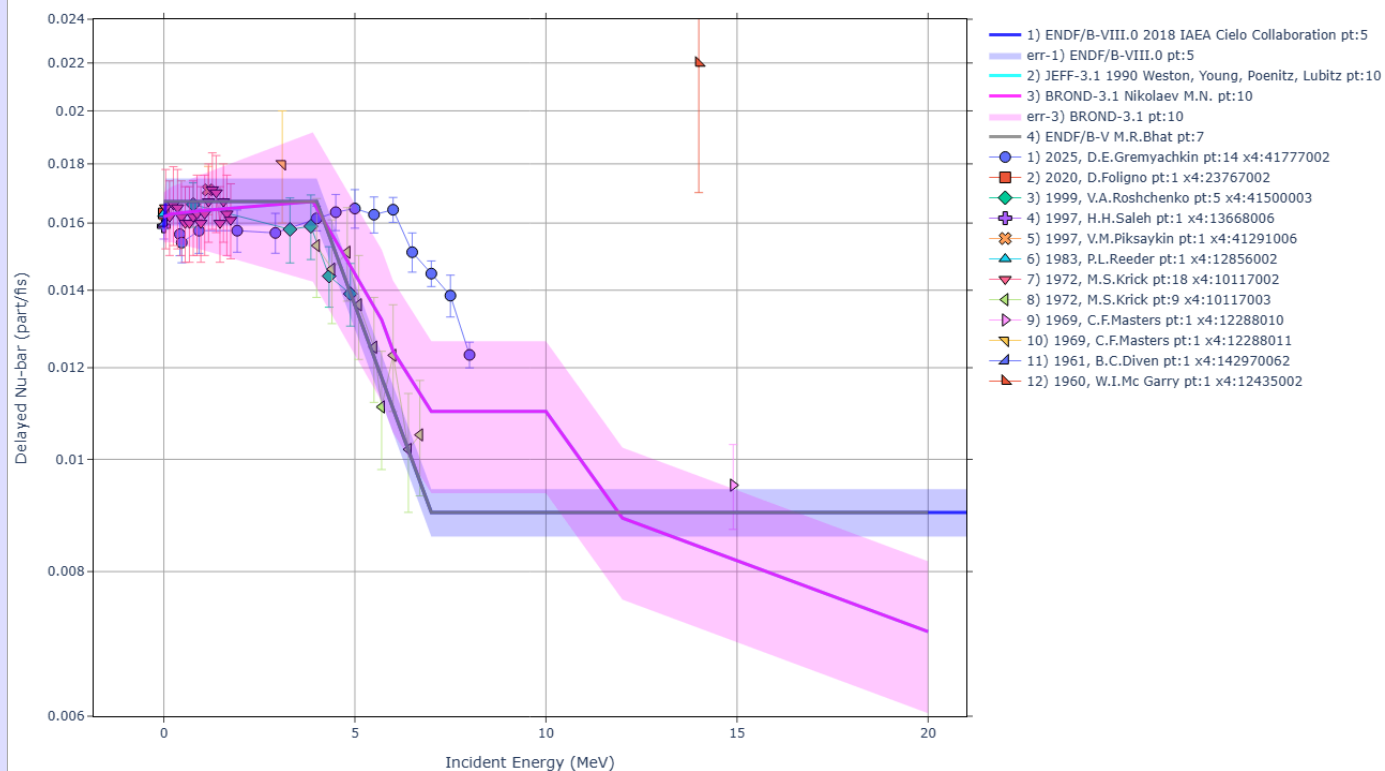


9) part2-9-nubar/out00/U235nf_pr-nu.html.png Prompt neutron yield (nu-bar)



10) part2-9-nubar/out00/U235nf_dl-nu.html.png Delayed neutron yield

EXFOR-ENDF NUBAR: 92-U-235(N,F),DL,NU Exp.Datasets:12 Points:54
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-08-25 23:14:02



Created by V.Zerkin, 09-Aug-2026
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